



(12) **United States Patent**
Brower

(10) **Patent No.:** **US 12,413,441 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **BLOCKCHAIN-BASED IoT**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/620,042**

(22) Filed: **Mar. 28, 2024**

(65) **Prior Publication Data**

US 2024/0340194 A1 Oct. 10, 2024

Related U.S. Application Data

(60) Provisional application No. 63/457,560, filed on Apr. 6, 2023.

(51) **Int. Cl.**
H04L 12/28 (2006.01)
H04L 67/10 (2022.01)
H04L 67/306 (2022.01)

(52) **U.S. Cl.**
CPC **H04L 12/2807** (2013.01); **H04L 67/10** (2013.01); **H04L 67/306** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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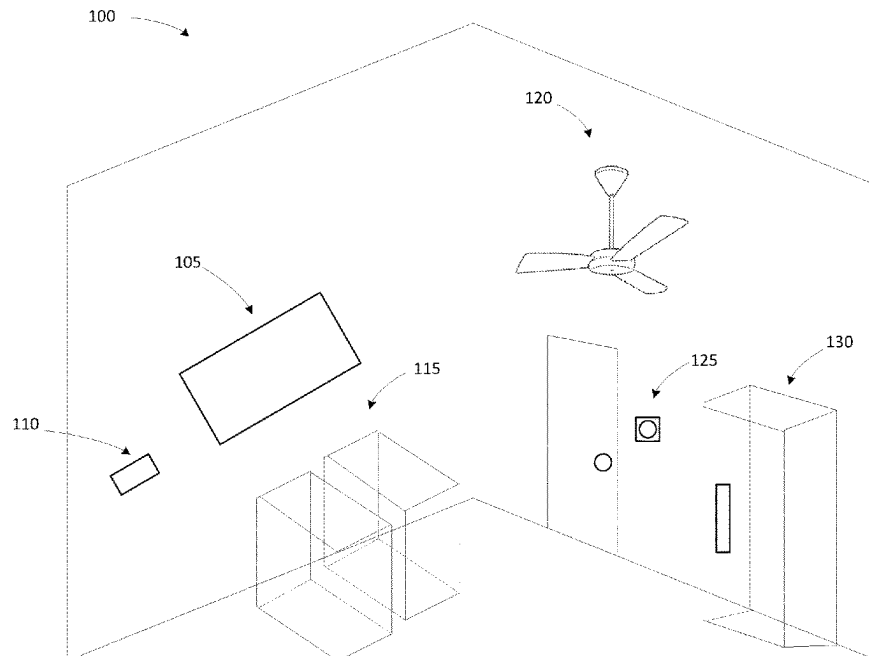
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(57) **ABSTRACT**

A method for user control of one or more devices located within a physical space. The method includes creating a network block in response to a user input, the network block including network identifying metadata. The network identifying metadata includes a list of the one or more devices. The method also includes writing the network block to a ledger under a first user profile, linking the network block to a host server, receiving a first request to modify the network block, and modifying the network block. Modifying the network block includes adding network identifying metadata, deleting network identifying metadata, and altering network identifying metadata. The method also includes receiving a second request to transfer ownership of the network block and linking the network block to a second
(Continued)



user profile. Linking the network block to the second user profile transfers ownership of the list of the one or more devices.

20 Claims, 8 Drawing Sheets

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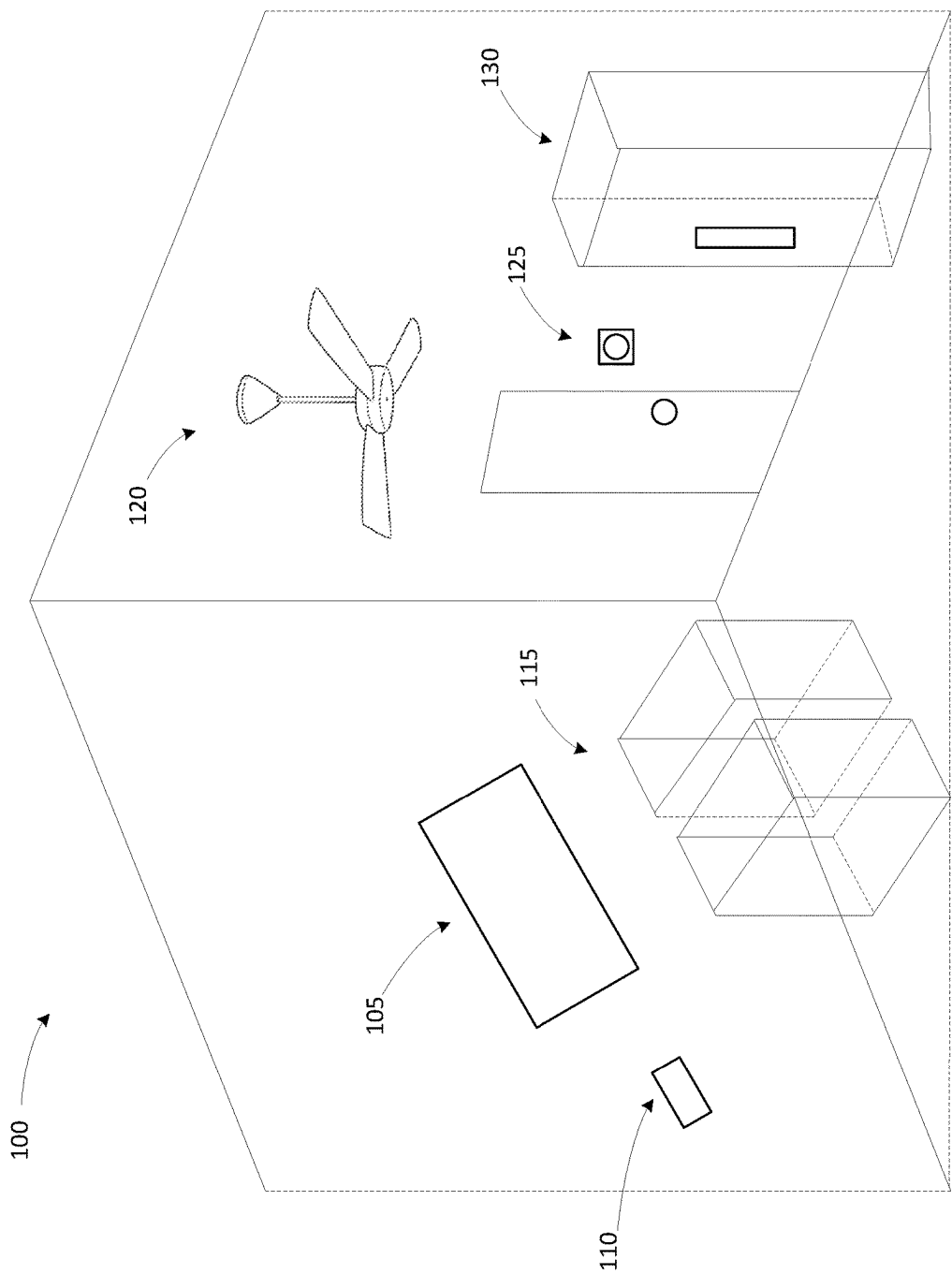


FIG. 1

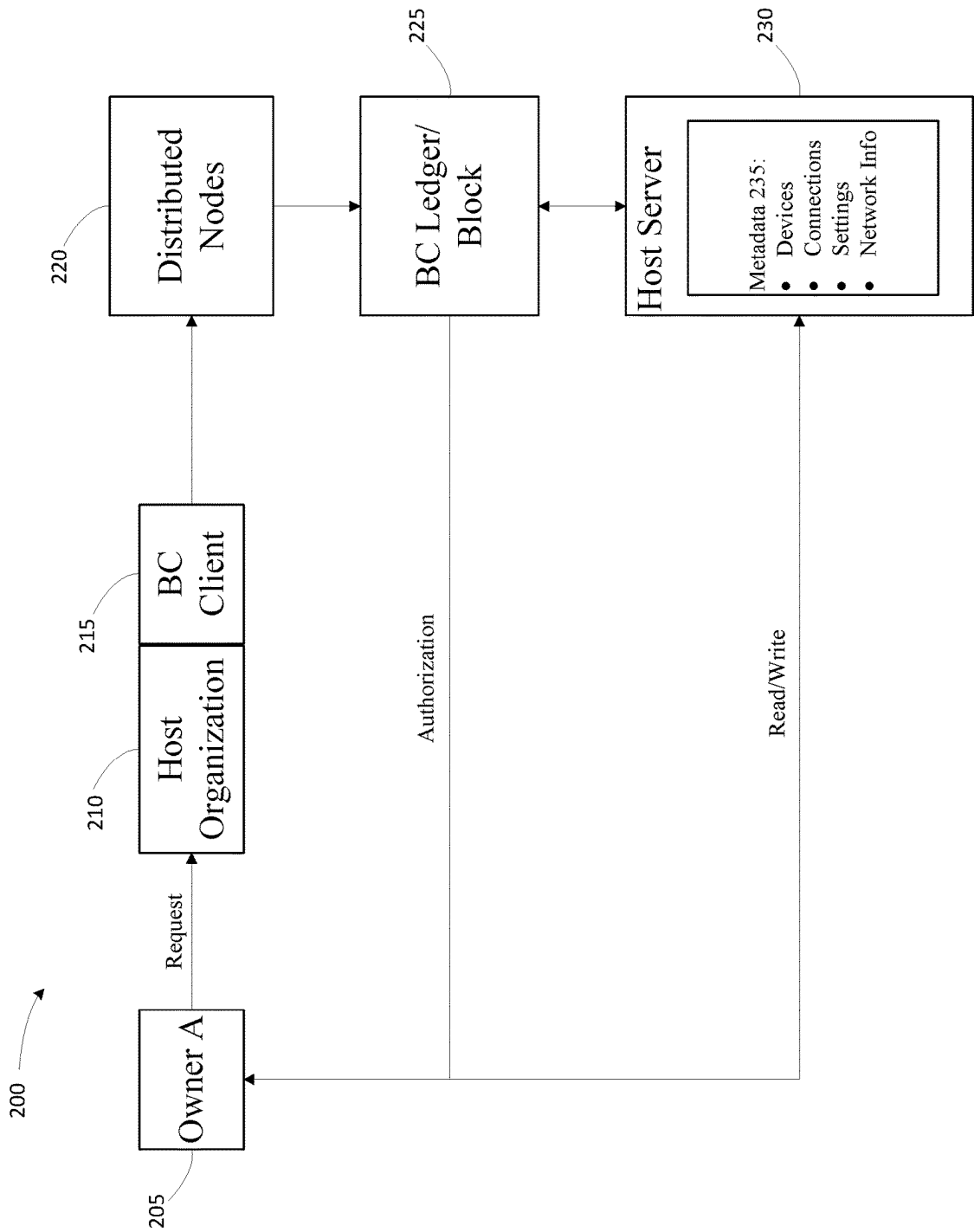


FIG. 2

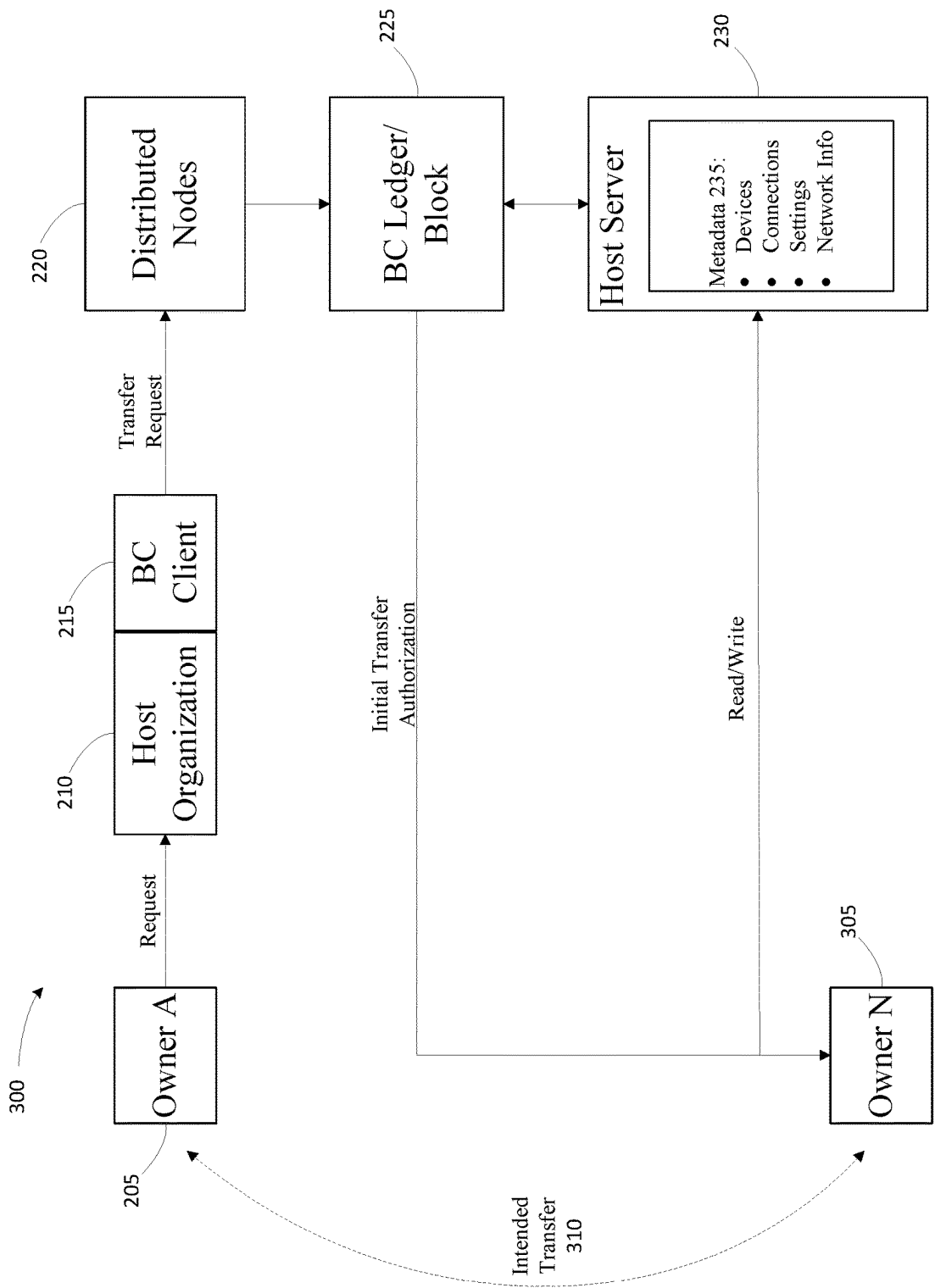


FIG. 3

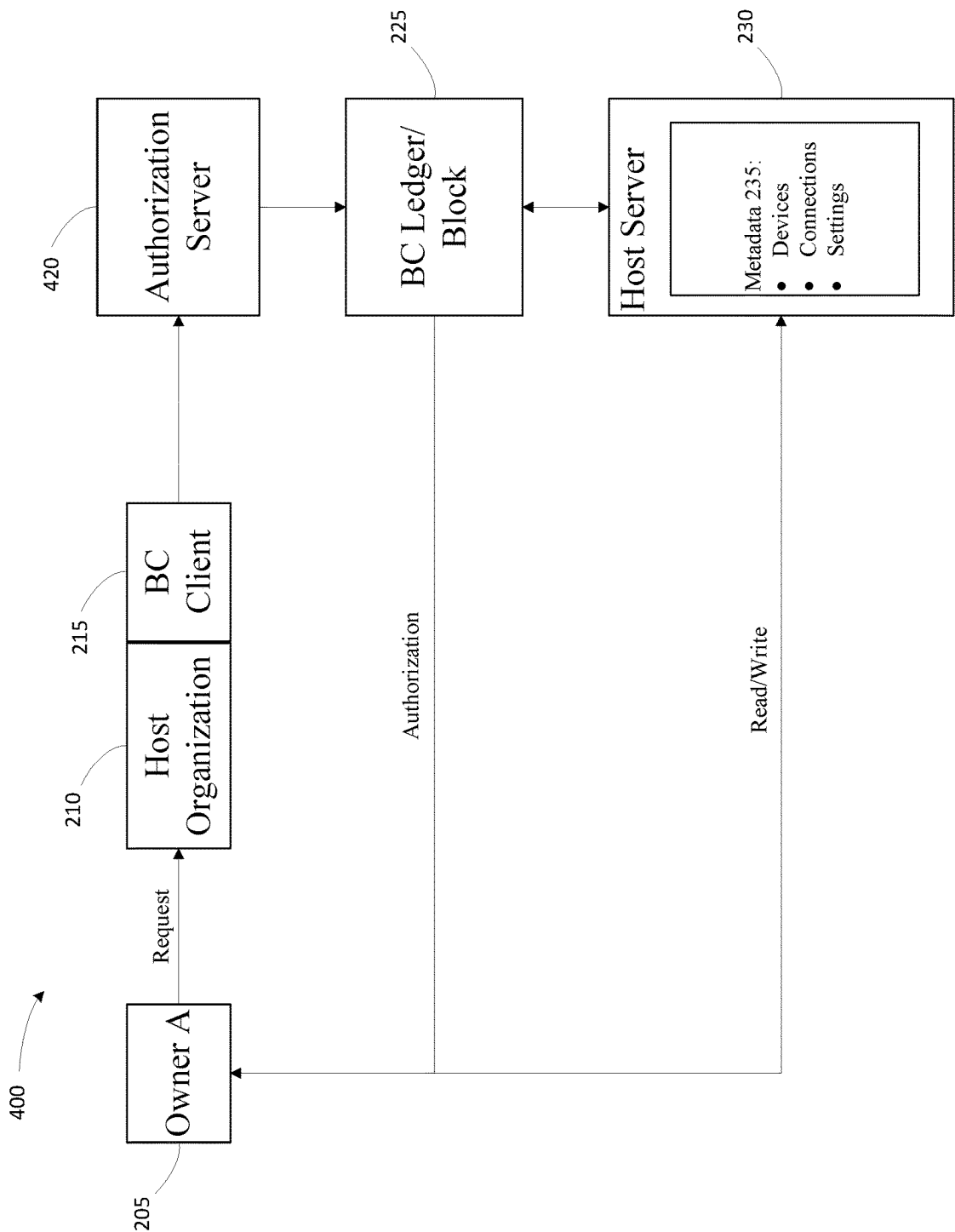


FIG. 4

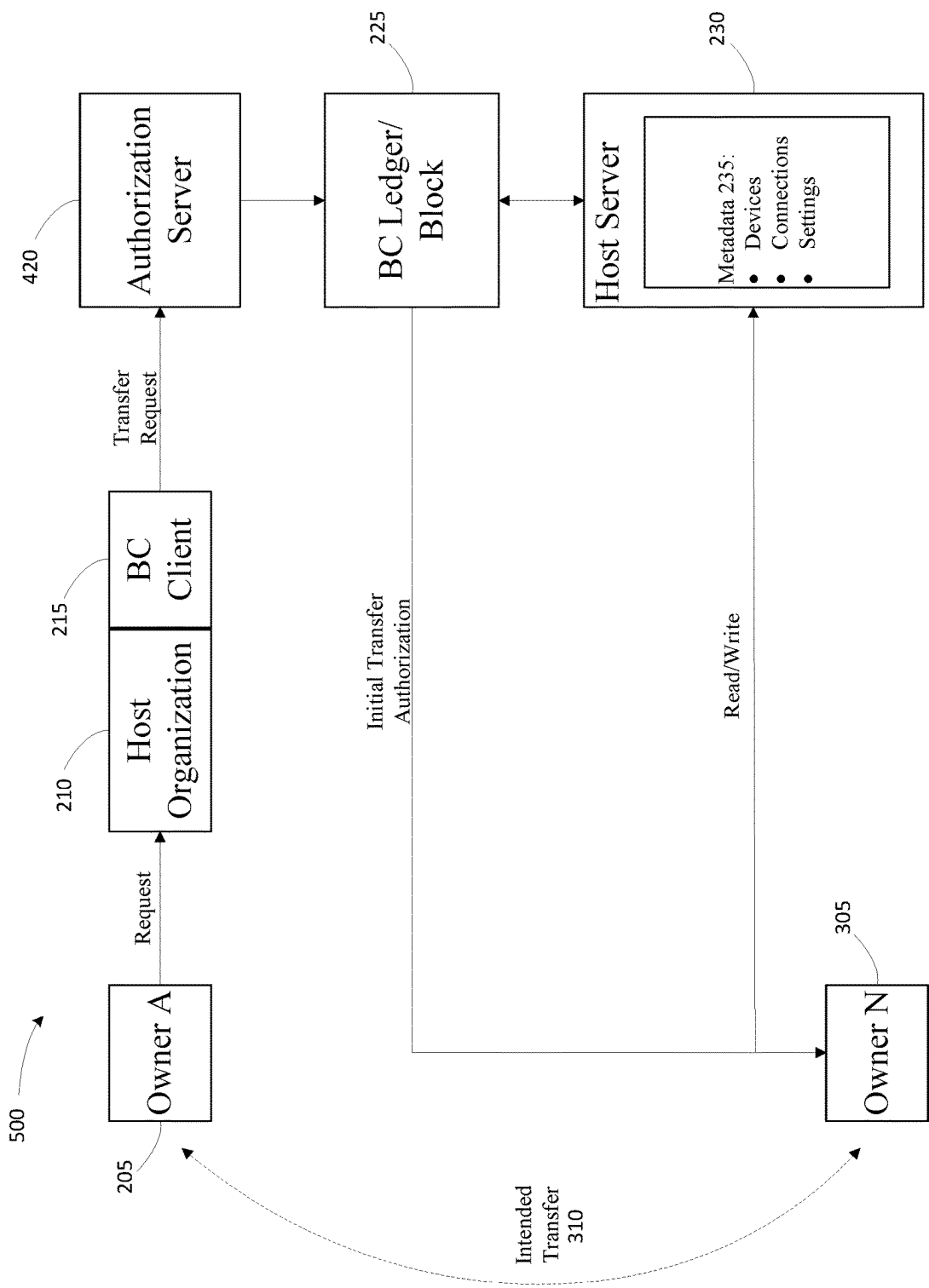


FIG. 5

FIG. 6

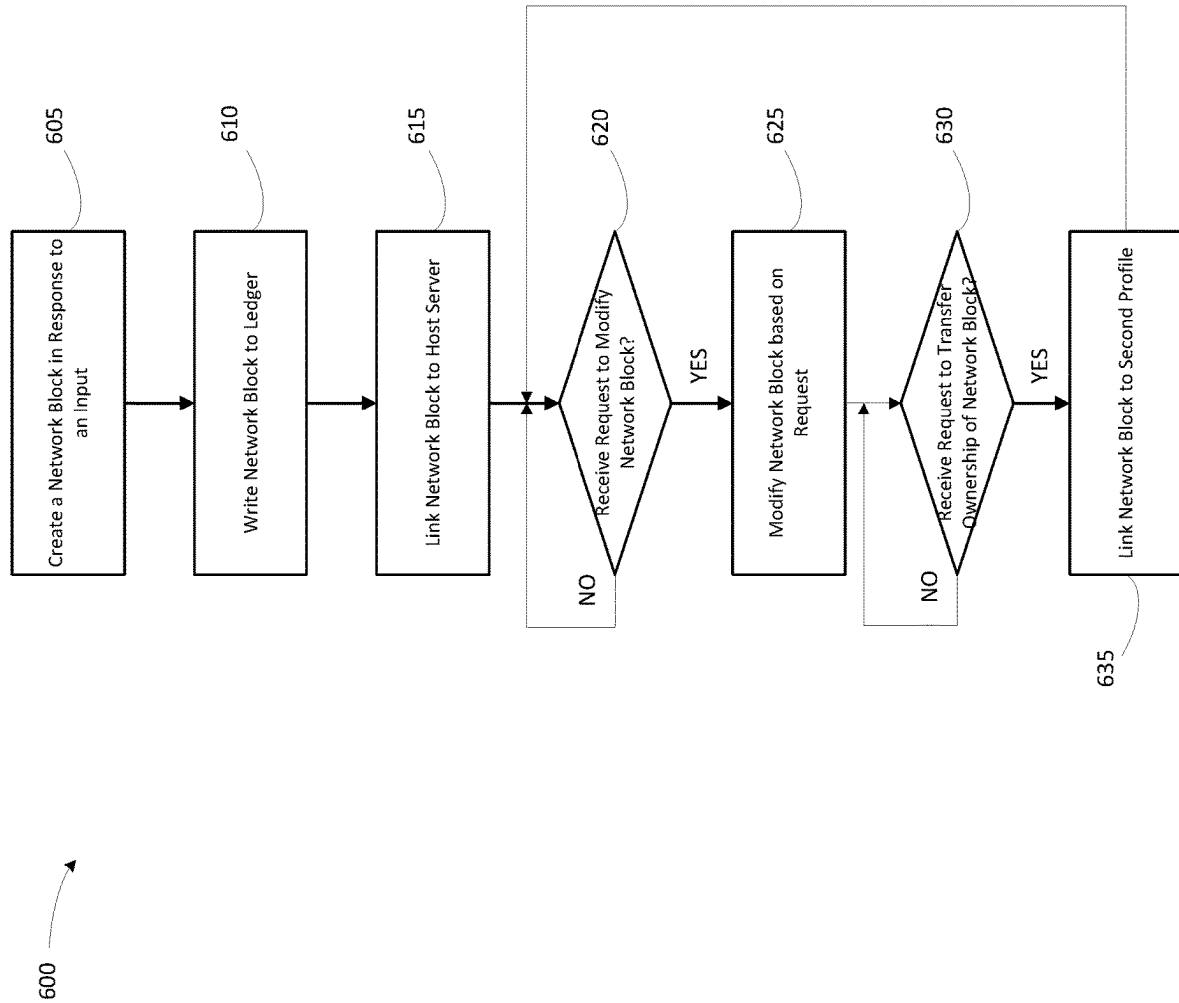


FIG. 7

700

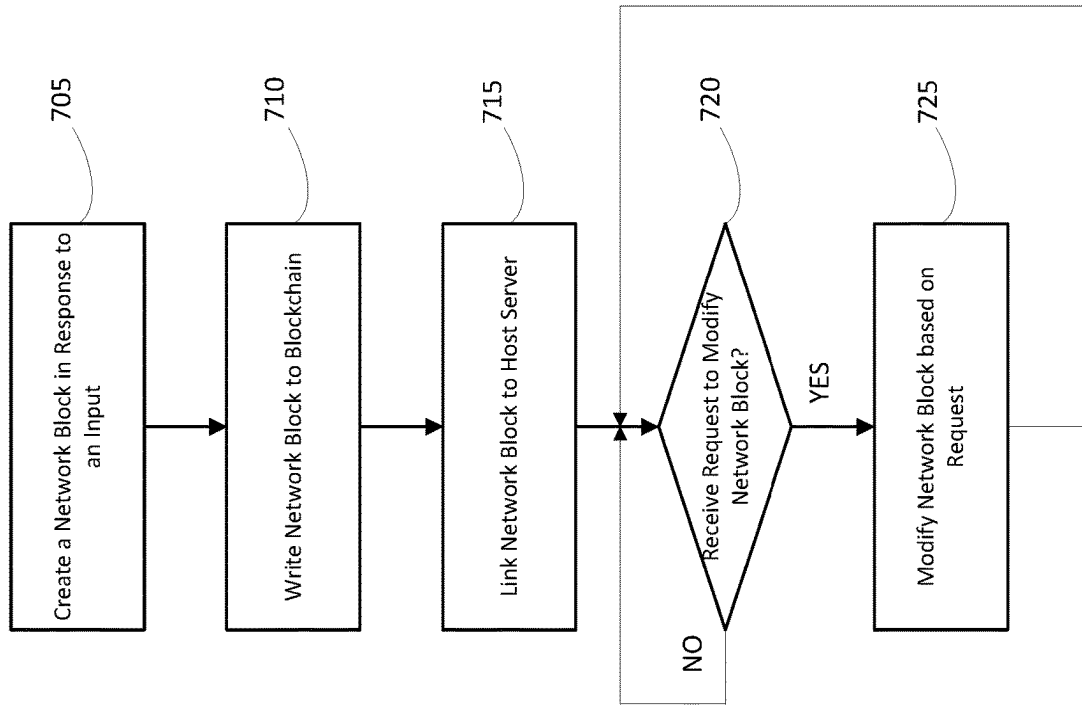
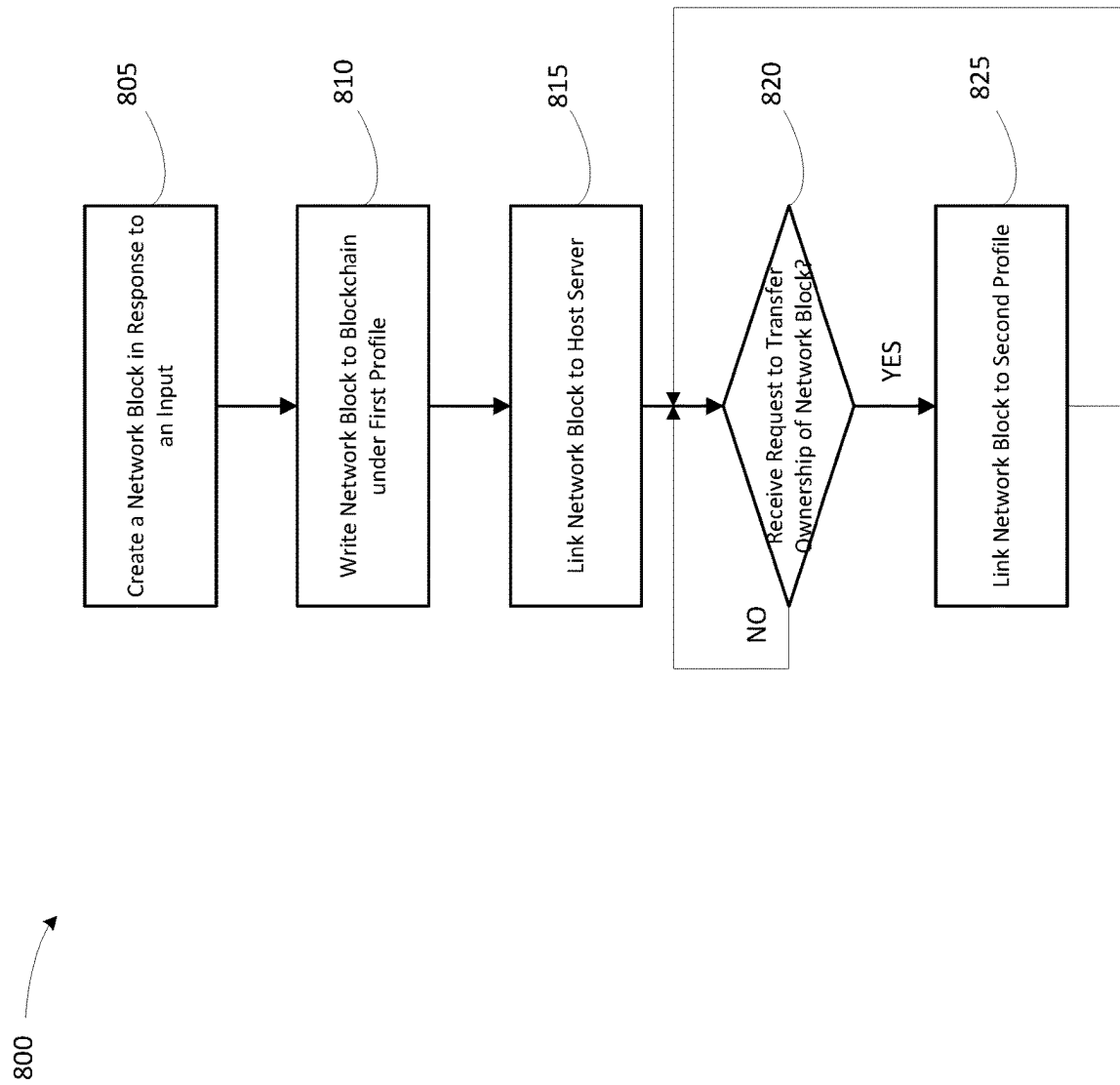


FIG. 8

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BLOCKCHAIN-BASED IoT**RELATED APPLICATIONS**

The present application claims priority to provisional patent application No. 63/457,560, which was filed on Apr. 6, 2023, the entire contents of which are hereby incorporated by reference.

FIELD

The embodiments disclosed herein relate to methods for user control of one or more devices located within a physical space.

SUMMARY

Modern living and working spaces may include one or more smart devices (e.g., fans, lighting receptacles, television, thermostats, etc.) and control the one or more devices via a network. Smart devices may be remotely controlled by an external device (e.g., a smartphone, tablet, etc.) in electronic communication with the network. Smart devices may also be configured to operate based on established user control settings for a specific living or working space. Wherein users may be owners, occupants, and/or tenants of the specific living or working space. Oftentimes, user control of living or working spaces changes, with the established user control settings remaining in effect after the change in user control of the living or working space occurs. New users may wish to modify or transfer the smart devices of the living or working space without reconfiguring every smart device on the network. In some instances, it may be difficult and time-consuming for new users to re-configure the specifications or settings for every smart device. Methods disclosed herein may allow for faster transfer of control of the smart devices from the established user to the new user. For example, control of one or more smart devices within the specific living or working space is transferred to the new user. The new user may access the smart devices without the need to reconfigure each and every smart device to the network with new user control settings.

The disclosure provides, in one aspect, a method for user control of one or more devices located within a physical space. The method includes creating a network block in response to a user input, the network block including network identifying metadata, and the network identifying metadata includes a list of the one or more devices located within the physical space. The method also includes writing the network block to a ledger under a first user profile, linking the network block to a host server, receiving a first request to modify the network block, and modifying the network block in response to receiving the first request. Modifying the network block includes at least one selected from the group consisting of adding network identifying metadata to the network block, deleting network identifying metadata from the network block, and altering the network identifying metadata within the network block. The method also includes receiving a second request to transfer ownership of the network block and linking the network block to a second user profile in response to receiving the second request. Linking the network block to the second user profile transfers ownership of the list of the one or more devices located within the physical space to the second user profile.

In some aspects, the list of the one or more devices includes at least one selected from the group consisting of a first set of devices, a second set of devices, and a third set

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of devices, the first set of devices being different than the second set of devices and the third set of devices, and the second set of devices being different than the third set of devices.

In some aspects, the first set of devices includes first identifying metadata, the second set of devices includes second identifying metadata, and the third set of devices includes third identifying metadata.

In some aspects, linking the network block to the second user profile transfers ownership of the first set of devices, the second set of devices, and the third set of devices to the second user profile.

In some aspects, writing the network block to the ledger includes storing the network block in a blockchain.

In some aspects, the list of the one or more devices includes smart home devices.

In some aspects, the method includes receiving a third request to create the first user profile, creating the first user profile based on the third request, receiving a fourth request to create the second user profile, and creating the second user profile based on the fourth request. The third request is created by a different user than the fourth request.

The disclosure provides, in another aspect, a method for modifying user control of one or more devices located within a physical space. The method includes creating a network block in response to a user input, the network block including network identifying metadata, and the network identifying metadata includes a list of the one or more devices located within the physical space. The method also includes writing the network block to a ledger under a first user profile, linking the network block to a host server, receiving a request to modify the network block, and modifying the network block in response to receiving the request to modify the network block.

In some aspects, the list of the one or more devices includes at least one selected from the group consisting of a first set of devices, a second set of devices, and a third set of devices, the first set of devices being different than the second set of devices and the third set of devices, and the second set of devices being different than the third set of devices.

In some aspects, the first set of devices includes first identifying metadata, the second set of devices includes second identifying metadata, and the third set of devices includes third identifying metadata.

In some aspects, modifying the network block includes adding network identifying metadata to the network block based on receiving the request to modify the network block.

In some aspects, modifying the network block includes deleting network identifying metadata from the network block based on receiving the request to modify the network block.

In some aspects, modifying the network block includes altering the network identifying metadata within the network block based on receiving the request to modify the network block.

The disclosure provides, in another aspect, a method for tracking user control of one or more devices located within a physical space. The method includes creating a network block in response to a user input, the network block including network identifying metadata, and the network identifying metadata includes a list of the one or more devices located within the physical space. The method also includes writing the network block to a ledger under a first user profile, linking the network block to a host server, receiving a first request to transfer ownership of the network block,

and linking the network block to a second user profile in response to receiving the first request to transfer ownership.

In some aspects, the list of the one or more devices includes at least one selected from the group consisting of a first set of devices, a second set of devices, and a third set of devices, the first set of devices being different than the second set of devices and the third set of devices, and the second set of devices being different than the third set of devices.

In some aspects, the first set of devices includes first identifying metadata, the second set of devices includes second identifying metadata, and the third set of devices includes third identifying metadata.

In some aspects, linking the network block to the second user profile transfers ownership of the first set of devices, the second set of devices, and the third set of devices to the second user profile.

In some aspects, the method includes storing the network block in a blockchain.

In some aspects, the list of the one or more devices includes the devices of the physical space.

In some aspects, the method includes receiving a second request to create the first user profile, creating the first user profile based on the second request, receiving a third request to create the second user profile, and creating the second user profile based on the third request. The second request is created by a different user than the third request.

Other aspects of the technology will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a physical space including one or more devices, in accordance with an embodiment of the disclosure.

FIG. 2 illustrates a system for user control of the one or more devices of FIG. 1, in accordance with an embodiment of the disclosure.

FIG. 3 illustrates a system for tracking user control of the one or more devices of FIG. 1, in accordance with an embodiment of the disclosure.

FIG. 4 illustrates a system for user control of the one or more devices of FIG. 1, in accordance with an embodiment of the disclosure.

FIG. 5 illustrates a system for tracking user control of the one or more devices of FIG. 1, in accordance with an embodiment of the disclosure.

FIG. 6 is a flow chart of a method for user control of one or more devices located within a physical space, in accordance with an embodiment of the disclosure.

FIG. 7 is a flow chart of a method for modifying user control of one or more devices located within a physical space, in accordance with an embodiment of the disclosure.

FIG. 8 is a flow chart of a method for tracking user control of one or more devices located within a physical space, in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION

Before any embodiments of the application are explained in detail, it is to be understood that the application is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. For ease of description, some or all of the example systems presented herein are illustrated with a single exemplar of each of its

component parts. Some examples may not describe or illustrate all components of the systems. Other exemplary embodiments may include more or fewer of each of the illustrated components, may combine some components, or may include additional or alternative components. The application is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof are meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

In addition, it should be understood that embodiments may include hardware, software, and electronic components or modules that, for purposes of discussion, may be illustrated and described as if the majority of the components were implemented solely in hardware. However, one of ordinary skill in the art, and based on a reading of this detailed description, would recognize that, in at least one embodiment, the electronic-based aspects may be implemented in software (e.g., stored on non-transitory computer-readable medium) executable by one or more electronic processors, such as a microprocessor and/or application specific integrated circuits (“ASICs”). As such, it should be noted that a plurality of hardware and software-based devices, as well as a plurality of different structural components, may be utilized to implement the embodiments. For example, “servers,” “computing devices,” “controllers,” “processors,” etc., described in the specification can include one or more electronic processors, one or more computer-readable medium modules, one or more input/output interfaces, and various connections (e.g., a system bus) connecting the components.

Methods described herein may be performed by one processor or a collection of processors located locally on a single device, or distributed among various local and remote locations (for example, in separate devices, located in the cloud, etc.). Additionally, all or some of the steps presented in each method may be used. In some embodiments, the steps are conducted in a different order.

FIG. 1 illustrates a physical space **100** including one or more devices, according to some embodiments. For example, the physical space **100** can be any enclosed living area or working area capable of supporting human inhabitants. In the illustrated embodiment, the physical space **100** is a smart home with the one or more devices being home appliances. The one or more devices may be electrically connected to and/or electronically communicating with a wireless network of the physical space **100**. In some embodiments, the one or more devices may be controlled remotely via the wireless network in response to one or more inputs from an external device.

As shown in FIG. 1, the one or more devices within the physical space **100** may include a first set of devices **105** (e.g., one or more smart TVs), a second set of devices **110** (e.g., one or more smart thermostats), a third set of devices **115** (e.g., a smart washer and/or dryer), a fourth set of devices **120** (e.g., a smart ceiling fan), a fifth set of devices **125** (e.g., a smart doorbell such as, but not limited to, doorbells with video and audio transmission capabilities), and a sixth set of devices (e.g., a smart fridge). It should be understood that the physical space **100** may include a greater or fewer number of sets of devices than shown. For example, the one or more devices may also include smart devices such as, but not limited to, water heaters, light receptacles, lighting fixtures, light bulbs, alarm/security systems, garage

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doors, audio equipment, dishwashers, automatic vacuums, etc. Although described above as a smart home, in some embodiments, the physical space **100** may be other living areas having different human inhabitants or owners over a period of time such as an apartment, a condo, a timeshare, etc. Although described as owner(s) below, transfers may occur between tenants, occupants, users, and/or the like.

In other embodiments, the physical space **100** is a larger, enclosed working area such as a hospital, an office building, a construction site, a grocery store, a warehouse, etc. In such embodiments, the one or more devices may include smart devices such as computers, industrial lighting fixtures, air conditioning systems, medical equipment, manufacturing equipment, etc. Additionally, when the one or more devices within the physical space **100** are configured to communicate with the wireless network, the one or more devices may be configured to certain specifications or settings based on a current user (e.g., an owner or tenant) of the physical space **100**. When ownership/occupancy of the physical space **100** is transferred to a new user (e.g., a new owner or new tenant), it may be difficult and time-consuming to re-configure the specifications or settings for each one of the one or more devices.

FIG. 2 illustrates a network system **200** for user control of the one or more devices within the physical space **100**, according to some embodiments. The embodiment shown in FIG. 2 may also be considered a blockchain ecosystem. In the illustrated embodiment, the network system **200** includes a first owner **205** (e.g., the current owner/user/tenant/occupant of the physical space **100**), a host organization **210**, a client **215**, a plurality of distributed nodes **220**, a ledger **225** (e.g., a blockchain ledger), and a host server **230**, which itself includes storage for network identifying metadata **235**. In some embodiments, the first owner **205** accesses the host organization **210** and the client **215** via a web application and/or a mobile application. In some embodiments, the web application and the mobile application are combined into a single application that performs the functions of both, as illustrated in FIG. 2 with the combination of the host organization **210** and the client **215**. The web application and the mobile application may be accessed using personal devices for example, but not limited to, a mobile device, a personal tablet, a personal computer, a company or organization computer, or the like. For example, the personal device may include an electronic processor (for example, a microprocessor, an electronic controller, a microcontroller, or another suitable programmable device), a memory, a first communication interface, and an input/output device.

The memory may be a non-transitory computer readable medium. The memory can include combinations of different types of memory, for example a ROM, a RAM (for example, DRAM, SDRAM, etc.), EEPROM, flash memory, a hard disk, an SD card, or other suitable magnetic, optical, physical, or electronic memory devices. The electronic processor is connected to the memory and executes software instructions that are capable of being stored in RAM of the memory (for example, during execution), ROM of the memory (for example, on a generally permanent basis), or another non-transitory computer readable medium for example another memory or a disc. The electronic processor is configured to execute, among other things, instructions related to operations and methods described herein.

The host organization **210** may be a third-party agency, company, and/or organization that provides the implementation of the client **215** via the web application or the mobile application. In some embodiments, the client **215** is a blockchain client. The client **215** may be software stored

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within the web application and/or the mobile application (or, for example, within the memory of the personal device or the memory of a remote server) that accesses and processes transactions. In some embodiments, the host organization **210** and the client **215** receive a user input (e.g., a request) from the first owner **205** indicative of creating a network block. The network block may be a virtualization of the one or more devices within the physical space **100** with a permanent link to the network identifying metadata **235**. In response to receiving the user input to create the network block, the client **215** may create the network block including the network identifying metadata **235** by communicating with the plurality of distributed nodes **220**. The plurality of distributed nodes **220** may be a collection of electronic processors within the network system **200** that implement the client **215**. For example, the plurality of distributed nodes **220** is a public blockchain in which each electronic processor of the collection of electronic processors is independently owned. In some examples, the plurality of distributed nodes **220** validates the network block at each node of the plurality of nodes **220**. When validating the network block, each node of the plurality of nodes **220** stores a copy of the network block for future validations. For example, the plurality of distributed nodes **220** stores the network block, including the network identifying metadata **235** in a shared, distributed digital ledger.

In some embodiments, the plurality of distributed nodes **220** are in electronic communication with a centralized, digital storage of a plurality of network blocks. In the illustrated embodiment, the plurality of distributed nodes **220** are in electronic communication with the ledger **225**. For example, the plurality of distributed nodes **220** post their stored transactions (e.g., creation of network blocks) to the ledger **225** for storage. In some embodiments, once the network block is stored in the ledger **225**, the ledger **225** encrypts the network block such that only the first owner **205** is able to view the network identifying metadata **235**. The ledger **225** may provide authorization to the first owner **205** to modify the network block, as described below. Additionally, the ledger **225** securely verifies the creation of the network block within the network system **200**.

In some embodiments, the ledger **225** is in electronic communication with the host server **230**. In some embodiments, the host server **230** facilitates interactions between the first owner **205** and the network block (including the network identifying metadata **235**) via the client **215**. For example, the host server **230** stores an encrypted copy of the network block. By accessing the client **215** via the web application or the mobile application, the first owner **205** may modify the network block. In some embodiments, each device of the one or more devices includes the network identifying metadata **235**. For example, the network identifying metadata **235** includes information associated with a list of the one or more devices within the network block, electrical connections of the one or more devices, settings or configurations of the one or more devices, and network system **200**.

FIG. 3 illustrates a network system **300** for tracking user control of the one or more devices within the physical space **100**, according to some embodiments. In the illustrated embodiment, the network system **300** may include similar components to the components of the network system **200**, as described above with reference to FIG. 2. In some embodiments, the network system **300** includes a second owner (user/tenant/occupant) **305**. For example, the second owner **305** is a subsequent owner/user/tenant/occupant of the physical space **100** after the first owner **205**. The first

owner **205** may transfer user control of the one or more devices of the physical space **100** to the second owner **305**. For example, the first owner **205** provides a user input (e.g., a transfer request) to the client **215** indicative of an intended transfer **310** of user control of the one or more devices to the second owner **305**. In some embodiments, the transfer request is provided via the web application and/or the mobile application. In other embodiments, the transfer request may be, or may be a result of, an electronic transfer of assets such as, but not limited to, a bank transfer, a bank transaction, a loan transfer, or the like for the physical space **100**. In some embodiments, when the physical space **100** transfers user control frequently (such as an apartment, a condo, a hospital room, etc.) for example, the transfer request may be provided to the second owner **305** via electronic mail from the web application or the mobile application. Upon receipt of the transfer request, the client **215** may process the transfer request as a transaction of user control of the network block from the first owner **205** to the second owner **210**. The client **215** may provide the transaction of user control to the plurality of distributed nodes **220**. The plurality of distributed nodes **220** may validate the transaction of the network block and store a copy of the network block with the second owner **305** having user control of the one or more devices. In some embodiments, the plurality of distributed nodes **220** post their stored transactions (e.g., the intended transfer **310** of the network block) to the ledger **225** for storage. In some embodiments, once the network block is stored in the ledger **225**, the ledger **225** encrypts the network block such that only the second owner **305** is able to view the network identifying metadata **235**. The ledger **225** may provide authorization to the second owner **225** to modify the network block, as described below. In some embodiments, the ledger **225** creates a link between the first owner **205** and the second owner **305** indicative of the transfer of user control of the one or more devices to the second owner **305**.

FIG. 4 illustrates a network system **400** for user control of the one or more devices within the physical space **100**, according to some embodiments. In the illustrated embodiment, the network system **400** may include similar components to the components of the network system **200**, as described above with reference to FIG. 2. In some embodiments, the network system **400** includes an authorization server **420**. The authorization server **420** may be a private blockchain. For example, the authorization server **420** is a server run by a third-party agency, company, or organization to validate the creation of the network block, as described above. When validating the network block, the authorization server **420** stores a copy of the network block for future validations. For example, the authorization server **420** stores the network block, including the network identifying metadata **235** in a private, distributed digital ledger. The authorization server **420** may provide greater security of the network block by setting permission levels or identification authorization such that only the first owner **205** can modify the network block.

In some embodiments, the authorization server **420** is in electronic communication with a centralized, digital storage of a plurality of network blocks. In the illustrated embodiment, the authorization server **420** is in electronic communication with the ledger **225**. For example, the authorization server **420** posts their stored transactions (e.g., creation of network blocks) to the ledger **225** for storage. In some embodiments, once the network block is stored in the ledger **225**, the ledger **225** encrypts the network block such that only the first owner **205** is able to view the network

identifying metadata **235**. The ledger **225** may provide authorization to the first owner **225** to modify the network block, as described below. Additionally, the ledger **225** securely verifies the creation of the network block within the network system **200**.

FIG. 5 illustrates a network system **500** for tracking user control of the one or more devices within the physical space **100**, according to some embodiments. In the illustrated embodiment, the network system **500** may include similar components to the components of the network system **400**, as described above with reference to FIG. 4. In some embodiments, the network system **400** includes the second owner **305**. In some embodiments, the first owner **205** provides the transfer request to the client **215** indicative of an intended transfer **310** of user control of the one or more devices to the second owner **305**. In some embodiments, the transfer request is provided via the web application or the mobile application. In other embodiments, the transfer request may be, or may be a result of, an electronic transfer of assets such as, but not limited to, a bank transfer, a bank transaction, a loan transfer, or the like for the physical space **100**. In some embodiments, for example when the physical space **100** transfers user control frequently (such as an apartment, a condo, a hospital room, etc.), the transfer request may be provided to the second owner **305** via electronic mail from the web application or the mobile application. Upon receipt of the transfer request, the client **215** may process the transfer request as a transaction of user control of the network block from the first owner **205** to the second owner **210**. The client **215** may provide the transaction of user control to the authorization server **420**. The authorization server **420** may validate the transaction of the network block and store a copy of the network block with the second owner **305** having user control of the one or more devices. In some embodiments, the authorization server **420** posts their stored transactions (e.g., the intended transfer **310** of the network block) to the ledger **225** for storage. In some embodiments, once the network block is stored in the ledger **225**, the ledger **225** encrypts the network block such that only the second owner **305** is able to view the network identifying metadata **235**. The ledger **225** may provide authorization to the second owner **225** to modify the network block, as described below. In some embodiments, the authorization server **420** provides authorization to the second owner **225**. In some embodiments, the ledger **225** creates a link between the first owner **205** and the second owner **305** indicative of the transfer of user control of the one or more devices to the second owner **305**.

FIG. 6 is a flow chart illustrating a method **600** for user control of one or more devices located within the physical space **100**, according to some embodiments. It should be understood that the order of the steps disclosed in the method **600** could vary. For example, additional steps may be added to the process and not all of the steps may be required, or steps shown in one order may occur in a second order. The method **600** begins at step **605** when the client **215** creates the network block in response to the user input indicative of creating the network block from the first owner **205**. In some embodiments, the network block includes the network identifying metadata **235**. The network identifying metadata **235** may include the list of the one or more devices located within the physical space **100**. For example, the list of the one or more devices includes the first set of devices **105**, the second set of devices **110**, and the third set of devices **115**. In some embodiments, the first set of devices **105** includes first identifying metadata, the second set of devices **110** includes second identifying metadata, and the

third set of devices **115** includes third identifying metadata. The first identifying metadata being different than the second identifying metadata and the third identifying metadata and the second identifying metadata being different than the third identifying metadata. It should be understood that the list of the one or more devices may include a greater or fewer number of devices than the number described above. The method **600** then proceeds to step **610**.

At step **610**, the client **215** writes the network block to the ledger **225** under a first user profile. For example, the client **215** provides the network block to the ledger **225** via the plurality of distributed nodes **220** or the authorization server **420**. The ledger **225** may store the network block under the first user profile associated with the first owner **205**. In some embodiments, the network block is stored in a blockchain of the ledger **225**. The method **600** then proceeds to step **615**. At step **615**, the ledger **225** links the network block to the host server **230**. For example, the ledger **225** provides the network block to the host server **230** and the host server **230** stores an encrypted copy of the network block for access by the first owner **205** via the client **215**. The method **600** then proceeds to step **620**.

At step **620**, the host server **230** determines whether a first request is received at the host server **230** to modify the network block. For example, the first owner **205** provides the first request to modify the network block to the host server **230** via the client **215**. When the host server **230** determines that the first request is not received, the method **600** may repeat step **620** until the host server **230** receives the first request. When the host server **230** determines that the first request is received, the method **600** then proceeds to step **625**. At step **625**, the host server **230** modifies the network block in response to receiving the first request. For example, the host server **230** adds the fourth set of devices **120** including fourth identifying metadata to the network block in response to receiving the first request. In other embodiments, the host server **230** deletes one or more devices within the physical space **100** from the network block. For example, the host server **230** deletes the third set of devices **115** including the third identifying metadata from the network block in response to receiving the first request. In other embodiments, the host server **230** alters the network identifying metadata **235** within the network block in response to receiving the first request. Once the network block is modified by the host server **230** provides the modified network block to the ledger **225**. The ledger **225** may store a copy of the modified network block. The method **600** then proceeds to step **630**.

At step **630**, the host server **230** determines whether a second request is received at the host server **230** to transfer ownership of the network block. For example, the first owner **205** provides the second request to transfer ownership of the network block to the second owner **305** via the client **215**. When the host server **230** determines that the second request is not received, the method **600** may repeat step **630** until the host server **230** receives the second request. When the host server **230** determines that the second request is received, the method **600** then proceeds to step **635**. At step **635**, the ledger **225** links the network block to a second user profile in response to receiving the second request. The second user profile may be associated with the second owner **305**. For example, the ledger **225** receives the second request indicative of the intended transfer **310** of user control of the network block including the list of the one or more devices to the second owner **305** and links the network block to the second owner **305**. In some examples, linking the network block to the second user profile transfers ownership of the

list of the one or more devices located within the physical space **100** to the second user profile (e.g., to the second owner **305**). For example, ownership of the first set of devices **105** including the first identifying metadata, the second set of devices **110** including the second set of identifying metadata, and the third set of devices **115** including the third identifying metadata is transferred to the second user profile. In some embodiments, after the ledger **225** links the network block to the second user profile, the method **600** may return to step **620** to continue to determine whether a request to modify the network block is received.

In some embodiments, the host organization **210** and the client **215** receive a third request to create the first user profile from the first owner **205** via the web application and/or the mobile application. For example, the first user profile is a unique exchange profile such that any actions performed by the web application and/or the mobile application are associated with the first owner **205** performing the actions. Based on receiving the third request, the web application and/or mobile application creates the first user profile. The first user profile may include a public and private key pair for the host organization **210** to encrypt all transactions made via the client **215** and for viewing of the network block on the host server **230**. Accordingly, only those with the needed key may view the transaction and the contents of the network block. Similarly, the host organization **210** and the client **215** may receive a fourth request to create the second user profile from the second owner **305** via the web application and/or the mobile application. For example, the second user profile is a unique exchange profile such that any actions performed by the web application and/or the mobile application are associated with the second owner **305** performing the actions. Based on receiving the fourth request, the web application and/or mobile application creates the second user profile. The second user profile may include a public and private key pair for the host organization **210** to encrypt all transactions made via the client **215** and for viewing of the network block on the host server **230**.

FIG. 7 is a flow chart of a method **700** for modifying user control of one or more devices located within a physical space **100**, according to some embodiments. It should be understood that the order of the steps disclosed in the method **700** could vary. For example, additional steps may be added to the process and not all of the steps may be required, or steps shown in one order may occur in a second order. The method **700** begins at step **705** when the client **215** creates the network block in response to the user input indicative of creating the network block from the first owner **205**. In some embodiments, the network block includes the network identifying metadata **235**. The network identifying metadata **235** may include the list of the one or more devices located within the physical space **100**. For example, the list of the one or more devices includes the first set of devices **105**, the second set of devices **110**, and the third set of devices **115**. In some embodiments, the first set of devices **105** includes first identifying metadata, the second set of devices **110** includes second identifying metadata, and the third set of devices **115** includes third identifying metadata. The first identifying metadata being different than the second identifying metadata and the third identifying metadata and the second identifying metadata being different than the third identifying metadata. It should be understood that the list of the one or more devices may include a greater or fewer number of devices than the number described above. The method **700** then proceeds to step **710**.

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At step 710, the client 215 writes the network block to the ledger 225 under the first user profile. For example, the client 215 provides the network block to the ledger 225 via the plurality of distributed nodes 220 or the authorization server 420. The ledger 225 may store the network block under the first user profile associated with the first owner 205. In some embodiments, the network block is stored in a blockchain of the ledger 225. The method 700 then proceeds to step 715. At step 715, the ledger 225 links the network block to the host server 230. For example, the ledger 225 provides the network block to the host server 230 and the host server 230 stores an encrypted copy of the network block for access by the first owner 205 via the client 215. The method 700 then proceeds to step 720.

At step 720, the host server 230 determines whether a first request is received at the host server 230 to modify the network block. For example, the first owner 205 provides the first request to modify the network block to the host server 230 via the client 215. When the host server 230 determines that the first request is not received, the method 700 may repeat step 720 until the host server 230 receives the first request. When the host server 230 determines that the first request is received, the method 700 then proceeds to step 725. At step 725, the host server 230 modifies the network block in response to receiving the first request. For example, the host server 230 adds the fourth set of devices 120 including fourth identifying metadata to the network block in response to receiving the first request. In some instances, the host server 230 adds network identifying metadata 235 to the list of the one or more devices based on receiving the first request. In other embodiments, the host server 230 deletes one or more devices within the physical space 100 from the network block. For example, the host server 230 deletes the third set of devices 115 including the third identifying metadata from the network block in response to receiving the first request. In some embodiments, the host server 230 deletes network identifying metadata 235 from the list of the one or more devices based on receiving the first request. In other embodiments, the host server 230 alters the network identifying metadata 235 within the network block in response to receiving the first request. Once the network block is modified by the host server 230 provides the modified network block to the ledger 225. The ledger 225 may store a copy of the modified network block. In some embodiments, after the host server 230 modifies the network block, the method 700 may return to step 720 to continue to determine whether a request to modify the network block is received.

FIG. 8 is a flow chart of a method 800 for tracking user control of one or more devices located within the physical space 100, according to some embodiments. It should be understood that the order of the steps disclosed in the method 800 could vary. For example, additional steps may be added to the process and not all of the steps may be required, or steps shown in one order may occur in a second order. The method 800 begins at step 805 when the client 215 creates the network block in response to the user input indicative of creating the network block from the first owner 205. In some embodiments, the network block includes the network identifying metadata 235. The network identifying metadata 235 may include the list of the one or more devices located within the physical space 100. For example, the list of the one or more devices includes the first set of devices 105, the second set of devices 110, and the third set of devices 115. In some embodiments, the first set of devices 105 includes first identifying metadata, the second set of devices 110 includes second identifying metadata, and the

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third set of devices 115 includes third identifying metadata. The first identifying metadata being different than the second identifying metadata and the third identifying metadata and the second identifying metadata being different than the third identifying metadata. It should be understood that the list of the one or more devices may include a greater or fewer number of devices than the number described above. The method 800 then proceeds to step 810.

At step 810, the client 215 writes the network block to the ledger 225 under the first user profile. For example, the client 215 provides the network block to the ledger 225 via the plurality of distributed nodes 220 or the authorization server 420. The ledger 225 may store the network block under the first user profile associated with the first owner 205. In some embodiments, the network block is stored in a blockchain of the ledger 225. The method 800 then proceeds to step 815. At step 815, the ledger 225 links the network block to the host server 230. For example, the ledger 225 provides the network block to the host server 230 and the host server 230 stores an encrypted copy of the network block for access by the first owner 205 via the client 215. The method 800 then proceeds to step 820.

At step 820, the host server 230 determines whether a first request is received at the host server 230 to transfer ownership of the network block. For example, the first owner 205 provides the first request to transfer ownership of the network block to the second owner 305 via the client 215. When the host server 230 determines that the second request is not received, the method 800 may repeat step 820 until the host server 230 receives the second request. When the host server 230 determines that the second request is received, the method 800 then proceeds to step 825. At step 825, the ledger 225 links the network block to a second user profile in response to receiving the first request. The second user profile may be associated with the second owner 305. For example, the ledger 225 receives the first request indicative of the intended transfer 310 of user control of the network block including the list of the one or more devices to the second owner 305 and links the network block to the second owner 305. In some examples, linking the network block to the second user profile transfers ownership of the list of the one or more devices located within the physical space 100 to the second user profile (e.g., to the second owner 305). For example, ownership of the first set of devices 105 including the first identifying metadata, the second set of devices 110 including the second set of identifying metadata, and the third set of devices 115 including the third identifying metadata is transferred to the second user profile. In some embodiments, after the ledger 225 links the network block to the second user profile, the method 800 may return to step 820 to continue to determine whether a request to transfer ownership of the network block is received.

In some embodiments, the host organization 210 and the client 215 receive a second request to create the first user profile from the first owner 205 via the web application and/or the mobile application, as described above. Based on receiving the second request, the web application and/or mobile application creates the first user profile. Similarly, the host organization 210 and the client 215 may receive a third request to create the second user profile from the second owner 305 via the web application and/or the mobile application, as described above. Based on receiving the third request, the web application and/or mobile application creates the second user profile.

Thus, the application provides, among other things, methods for user control of one or more devices located within a

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physical space. Various features and advantages of the application are set forth in the following claims.

What is claimed is:

1. A method for user control of one or more devices located within a physical space, the method comprising:

creating a network block in response to a user input, the network block including network identifying metadata, wherein the network identifying metadata includes a list of the one or more devices located within the physical space;

writing the network block to a ledger under a first user profile;

linking the network block to a host server;

receiving a first request to modify the network block;

modifying the network block in response to receiving the first request,

wherein modifying the network block includes at least one selected from a group consisting of adding network identifying metadata to the network block, deleting network identifying metadata from the network block, and altering the network identifying metadata within the network block;

receiving a second request to transfer ownership of the network block; and

linking the network block to a second user profile in response to receiving the second request,

wherein linking the network block to the second user profile transfers ownership of the list of the one or more devices located within the physical space to the second user profile.

2. The method of claim 1, wherein the list of the one or more devices includes at least one selected from the group consisting of a first set of devices, a second set of devices, and a third set of devices, the first set of devices being different than the second set of devices and the third set of devices, and the second set of devices being different than the third set of devices.

3. The method of claim 2, wherein the first set of devices includes first identifying metadata, the second set of devices includes second identifying metadata, and the third set of devices includes third identifying metadata.

4. The method of claim 3, wherein linking the network block to the second user profile transfers ownership of the first set of devices, the second set of devices, and the third set of devices to the second user profile.

5. The method of claim 1, wherein writing the network block to the ledger includes storing the network block in a blockchain.

6. The method of claim 1, wherein the list of the one or more devices includes smart home devices.

7. The method of claim 1, further comprising:

receiving a third request to create the first user profile; creating the first user profile based on the third request; receiving a fourth request to create the second user profile; and

creating the second user profile based on the fourth request,

wherein the third request is created by a different user than the fourth request.

8. A method for modifying user control of one or more devices located within a physical space, the method comprising:

creating a network block in response to a user input, the network block including network identifying metadata, wherein the network identifying metadata includes a list of the one or more devices located within the physical space;

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writing the network block to a ledger under a first user profile;

linking the network block to a host server;

receiving a request to modify the network block; and

modifying the network block in response to receiving the request to modify the network block.

9. The method of claim 8, wherein the list of the one or more devices includes at least one selected from the group consisting of a first set of devices, a second set of devices, and a third set of devices, the first set of devices being different than the second set of devices and the third set of devices, and the second set of devices being different than the third set of devices.

10. The method of claim 9, wherein the first set of devices includes first identifying metadata, the second set of devices includes second identifying metadata, and the third set of devices includes third identifying metadata.

11. The method of claim 10, wherein modifying the network block further includes:

adding network identifying metadata to the network block based on receiving the request to modify the network block.

12. The method of claim 10, wherein modifying the network block further includes:

deleting network identifying metadata from the network block based on receiving the request to modify the network block.

13. The method of claim 10, wherein modifying the network block further includes:

altering the network identifying metadata within the network block based on receiving the request to modify the network block.

14. A method for tracking user control of one or more devices located within a physical space, the method comprising:

creating a network block in response to a user input, the network block including network identifying metadata, wherein the network identifying metadata includes a list of the one or more devices located within the physical space;

writing the network block to a ledger under a first user profile;

linking the network block to a host server;

receiving a first request to transfer ownership of the network block; and

linking the network block to a second user profile in response to receiving the first request to transfer ownership.

15. The method of claim 14, wherein the list of the one or more devices includes at least one selected from the group consisting of a first set of devices, a second set of devices, and a third set of devices, the first set of devices being different than the second set of devices and the third set of devices, and the second set of devices being different than the third set of devices.

16. The method of claim 15, wherein the first set of devices includes first identifying metadata, the second set of devices includes second identifying metadata, and the third set of devices includes third identifying metadata.

17. The method of claim 16, wherein linking the network block to the second user profile transfers ownership of the first set of devices, the second set of devices, and the third set of devices to the second user profile.

18. The method of claim 14, further comprising:

storing the network block in a blockchain.

19. The method of claim 14, wherein the list of the one or more devices includes the devices of the physical space.

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20. The method of claim 14, further comprising:
receiving a second request to create the first user profile;
creating the first user profile based on the second request;
receiving a third request to create the second user profile;
and
creating the second user profile based on the third request,
wherein the second request is created by a different user
than the third request.

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